

Sum coding:

two groups: A and B

$$\text{mean}(A) = 10$$

$$\text{mean}(B) = 14$$

$$\text{intercept} = (10 + 14) / 2 = 12 \rightarrow \text{grand mean}$$

$$\text{slope} = (10 - 14) / 2 = -2 \rightarrow \text{deviation of group}$$

$$Y = 12 - 2X$$

A from grand mean

Reconstruct the means.

$$\text{Group A} = \text{intercept} + \text{slope} = 12 + (-2) = 10$$

$$\text{Group B} = \text{intercept} - \text{slope} = 12 - (-2) = 14$$

because the slope estimate of the base

line group is $-(\beta_1 + \beta_2 + \dots + \beta_n)$.

the sum of all slopes = 0.

3 groups A, B, C

d_1 d_2

Coding - A 1 0

B 0 1

C -1 -1 $\rightarrow$ base

$$A_{\text{mean}} = \beta_0 + \beta_1 + 0$$

$$B_{\text{mean}} = \beta_0 + 0 + \beta_2$$

$$C_{\text{mean}} = \beta_0 - \beta_1 - \beta_2 \quad (\text{group C slope} = -(\beta_1 + \beta_2))$$